Generative Terrain Fast Prototyping in Virtual Reality with Freehand Sketching Interface

Yushen Hu
New York University
New York, NY, USA
yh3856@nyu.edu

Keru Wang
New York University
New York, NY, USA
keru.wang@nyu.edu

Zhu Wang
New York University
New York, NY, USA
zhu.wang@nyu.edu

Ken Perlin
New York University
New York, NY, USA
perlin@cs.nyu.edu

Abstract

Terrain generation and authoring in Virtual Reality (VR) offers unique benefits for terrain authoring including stereo display, immersive and intuitive design experience, and natural input modalities. We present this VR-based terrain fast prototyping system to integrate natural input modalities, preserve artistic controls and lower the effort of landscape prototyping. The system utilizes freehand interfaces and a generative model to help users quickly prototype different types of natural landscapes, such as mountains, mesas, canyons, and volcanoes. With the freehand interfaces, users can use their hands to draw mid-air strokes as the 3D contours of the desired landscapes. Then, a Conditional Generative Adversarial Network (CGAN) generates realistic landscapes based on the 3D contour. The freehand interface detects users' gestures to control landscape editing. By incorporating CGAN as the terrain synthesizer, our system allows users to immersively, rapidly, and easily prototype terrains using intuitive interactive hand controls.

Keywords

Virtual Reality, Terrain Generation, Hand Tracking

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1 Introduction

Terrain authoring covers the interactive process of creating and editing digital landscapes by simulating geographic features such as mountains, canyons, volcanoes, and mesas. Modern Virtual Reality (VR) platforms have advanced technical features such as hand-tracking and speech input in addition to stereo displays. They provide unique benefits for terrain authoring, particularly in the context of design, visualization, and interaction. In addition to the input modalities, VR can benefit perceptions during authoring terrains. Users can observe their models from their desired point of view by simply moving their head. This dynamic perspective shift enhances spatial awareness by improving users' perception of scale and spatial relationships between different segments. As a result, VR could potentially transform the traditional design process. They can immersively create, edit, navigate and experience virtual landscapes as if they were physically present, which can help to maintain a comprehensive understanding of the landscapes on the fly. Additionally, designing and building terrains in VR provides a unique benefit if the terrains are intended for a VR experience. In this case, designers can immerse themselves into their target user's perspective, which facilitates a more user-centered design approach.

Yet fast prototyping, terrain fidelity and user-friendly interfaces continue to pose challenges. Landscapes often have diverse features, making it inherently complex to generate realistic terrains while achieving the intended aesthetic style. To mitigate these issues, we present our VR-based terrain authoring system which, by incorporating CGAN as a generative terrain synthesizer, allows users to immersively, rapidly and easily prototype terrains using intuitive interactive hand controls.

Our system integrates the VR platform's build-in hand tracking and gesture recognition functions as its primary input modality. The hand tracking sub-system uses pinching as the action trigger and then tracks the pinch point for mid-air stroke drawing in VR. Similar to prior work [Coninx et al. 1997; Pfeuffer et al. 2017], we integrate pinch to facilitate direct freehand manipulations with rich controls such as pinch-to-select and two-handed scaling. Our exploration extends to the design space of the hand interface, which provides a comprehensive demonstration of all supported interactions across various terrain generation scenarios. A mid-air stroke, representing the 3D contours of the terrain, captures both horizontal layout and variations in the vertical dimension. This helps the system to efficiently generate terrains that match the user-specified height and shape, without the need for extensive adjustments to elevation required by most other terrain generation tools[Guérin et al. 2017, 2022]. The generative model for terrain generation is a CGAN module trained on pairs of real-world terrains and their corresponding height maps. The module synthesizes height information extracted from the user's mid-air strokes to generate a realistic landscape that reflects the user's creative intent and aligns with the patterns and features learned from the training dataset.

2 System Implementation

Our objective is to design data-driven terrain authoring tools in Virtual Reality (VR) for simple and fast 3D terrain prototyping. Our emphasis is on integrating natural 3D input modalities while preserving artistic control and lowering the effort required for landscape prototyping. We aim to create an intuitive and immersive terrain authorization experience for users.

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Based on this design goals, we propose a pinch-based VR terrain authoring system that harnesses its user's inherent understanding of three-dimensional space to create and manipulate 3D terrain. Our system allows users to create terrain using hand gestures to draw contours or filled polygons. The system then automatically generates realistic terrain based on these inputs. Users can further refine the terrain with pinching by editing its shape, position, and slope to achieve the desired landscape.

The system is built on Unity for its rich libraries in XR Interaction. For the construction and interaction with the terrain, we use the official terrain plugin from the Unity development team. For the machine learning integration, we use the Barracuda plugin of Unity.

2.1 Basic Setup

Each user wears a Meta Quest 3 headset with video passthrough capability. This setup allows users to see their physical surroundings augmented with superimposed 3D graphics, blending virtual and physical environments.

2.2 Gesture Detection

The system integrates the Oculus Interaction SDK package in Unity 3D with the built-in features in Meta Quest 3. Leveraging the built-in cameras on Meta Quest headsets, the SDK provides essential hand tracking functionality, including positional and orientation tracking of hands, configuration tracking of fingers, and hand gesture recognition.

2.3 Terrain Synthesis

The system will reconstruct the terrain from the contours that users have drawn, then, the terrain will be encoded into a 2D array as the input of the CGAN model for adding realistic details. After having the output from the CGAN model, it will be sent back to the Unity and be encoded into the terrain.

3 User Experience

The participants will play with our system through a single-player terrain generation process, which will take approximately 5-10 minutes. They will first watch a tutorial video and learn how to use the system by our instructor. This step will take around 2 minutes. Participants can practice the interaction controls in a practice session. After the tutorial, the participants will enter an empty scene for them to create and edit their desired landscape on the 3D terrain canvas. When participants finished creating the terrain, post-processing effects can be added to the terrain, such as landscape texture, procedurally-generated foliage, artificial buildings, and water bodies (rivers and lakes). For the landscape texture and foliage, we provide several themes including desert, tropical, and mountainous areas. Participants will be able to rearrange the position and orientation of the generated elements.

The shapes and characteristics of the terrain in our system are controlled by the 3D contours that participants sketch. The height difference of the contour will determine whether it represents a mountain or a valley. If the contour is above the terrain canvas which represents the ground level, the terrain will be extruded to form a mountain shape; but if it is under the terrain canvas, it will subside into a valley.

Terrain Creation. Users can pinch to start drawing contours in the mid-air, and release to end the drawing. Participants can only modify part of the terrain creation by editing part of the contours. As a result, the re-generation will be applied only to the edited part.

Terrain Editing. Users can pinch on existing terrain contours to modify them. By pinching a specific point on the contour using one hand, they can manipulate the position of that point causing the rest of the contour to adjust linearly. By pinching onto two points of the contour, they can move, rotate, and scale the part of the contour between those two points, and the rest of the contour will be affected linearly as it is in one-hand pinching. This feature allows users to edit the slope, ensuring the resulting terrain is realistic and reasonable. When pinching on one point of the contour, there will be two white lines coming from that point, representing the slope lines of the terrain contour. Users can pinch on the slope lines to adjust the shape of the slope and its steepness to create terrain features like cliffs.

Terrain Saving/Deletion. Saving terrain allows users to save it for revisit in the future, or make the existing terrain uneditable if they are happy with the generation. Users can save the entire existing terrain contours by clicking (rapidly pinch and release) with their left hand. And if they want to only save a particular contour, they can pinch that contour with their right hand, and perform the clicking action on their left hand.

If users want to delete the contour, they can pinch on that particular contour with their right hand to select it and click their left hand twice in a row to perform delete action. Clicking twice without pinching any contour will delete the entire existing contours and reset the terrain canvas.

Fill-mode Switch. Users can adjust the fill type of the contour, similar to common paint tools, to create various types of terrain. The fill mode can fill the inside of closed contours to create terrain like mesa and plateau. To turn on or off the fill mode, users can click once with their right hand. A paint bucket icon will appear when the fill mode is on.

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